

Introduction to Computing at Scale

The Design, Operation, and Societal Impacts of Data Centers

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Information & Computing

What is Information?

“Knowledge obtained from investigation, study, or instruction”

— Merriam-Webster

Digital

Binary data in memory

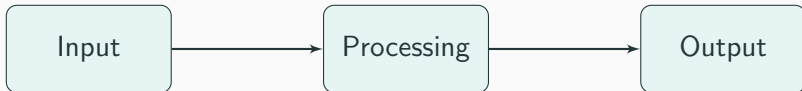
Algorithmic

Output of an algorithm

Physical

Heat from a stove

Information Processing is Valuable



- **Biology:** Hunger → Look for food → Eat → Not hungry.
- **Umwelt:** How different species experience the world.
- *Recommended Reading:* *An Immense World* by Ed Yong.

Standardizing and Math

Humans have intuition (more vs. less or nothing), but precision and collaboration are needed.

The Solution: MATH

Math serves as a common framework.

- **Standards** → Easier communication.
- Number systems, operations, written forms.

Computation = Information Processing + Math?

Why We Need Computers

The human brain is **bad at math**, and we aren't meant to be tools.

- **Limited Memory:** Working memory of 5–9 'numbers'
- **Unreliable:** Forgetful and non-operational for long periods of time (sleep)
- **Imprecise**
- **Hard to scale**
- **Slow:** Long training times (childhood/education).

We needed an optimized tool for the job.

Computing = Information Processing using Math?

Definitions:

- To determine especially by mathematical means
- To determine or calculate by means of a computer

Computing Hardware

How do we build fast, small, and efficient computers?

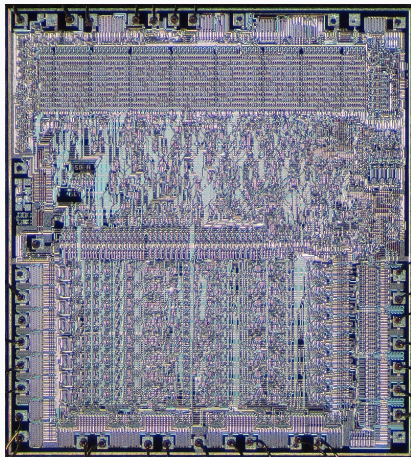
Currently and for many decades:

Transistors

- **Digital:** 0s and 1s.
- **Electrical:** Voltage levels.

Although this doesn't mean there aren't other alternatives

(see quantum, analog, neuro, etc).



Computing at Scale

Moore's Law: Transistor count per microprocessor doubles every ≈ 2 years.

Dennard Scaling:

- Power density stays constant as transistors shrink.
- Same chip area $\rightarrow$ Same power draw.

These trends drove decades of progress.

50 Years of Trends

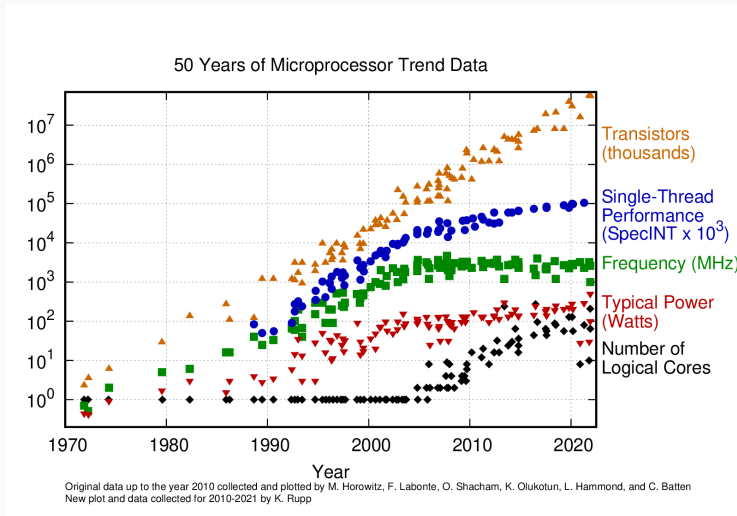
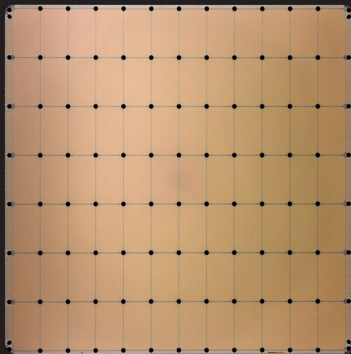


Figure 1: Transistors, Frequency, Power, and Cores (1970–2020)

Wafer-Scale Engines

Making bigger chips. From reticle-sized to wafer-sized.

Cerebras Wafer-Scale Engine



Cerebras WSE-2 7nm
2.6 Trillion Transistors
46,225 mm² Silicon

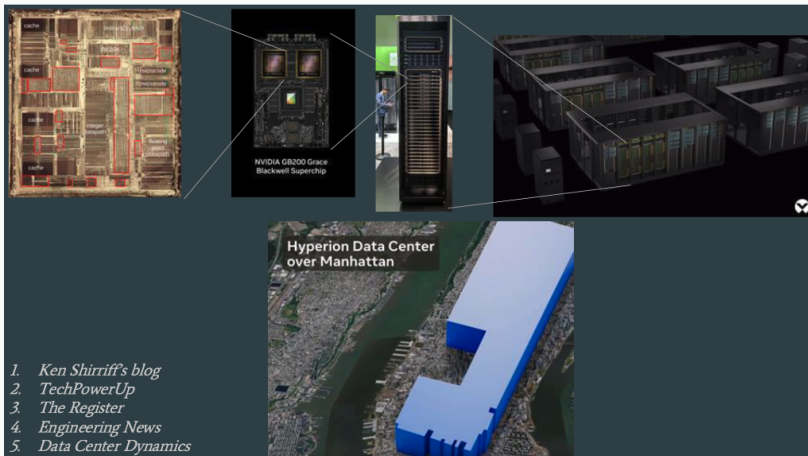


Largest GPU
54.2 Billion Transistors
826 mm² Silicon

Beyond the Chip

Scale isn't just on-chip. It's the whole system.

Transistors → Chips → Boards → Racks → Aisles → Datacenters.



The Scales of Computing

We scaled transistors **down**, and hit hard physical limits.

We scaled **up** and made bigger chips.

We connected more chips together and scaled **out**.

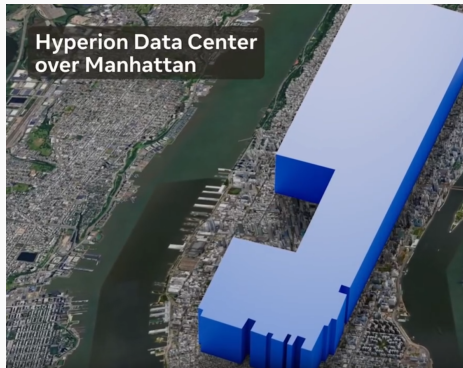
Now even a single datacenter is too small, and we scale **across**.

Impact: Power & Water

Unprecedented Energy Scale

The Jump in Scale

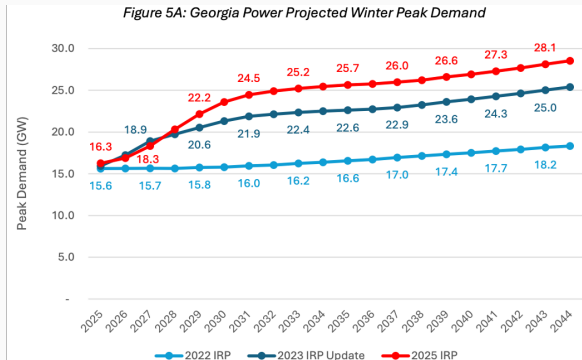
- **Past:** Top supercomputers ≈ 20 MW
- **Future:** Meta's "Hyperion" DC ≈ 5 GW.
- Note: A nuclear plant outputs ≈ 1 GW.



Hyperion Data Center scale over Manhattan

The Strain on our Energy Grid.

- Demand is outstripping generation capacity.
- Demand spiked **rapidly**. Decadal planning → ASAP.
- AI training loads can fluctuate 100s of MWs in milliseconds.
- Inflated demand through speculation and multiple bids
- AI bubble?



Source: Georgia Power IRP

The Water Cost

Massive Power + Inefficiencies = Cooling Demand

The Cost of Evaporative Cooling

- Evaporative cooling is very effective and power-efficient, but not without tradeoffs.
- A single DC *can* evaporate millions of gallons of water/day.
- Equivalent to the usage of hundreds of thousands of people.
- Can be mitigated by: free-air/dry/adiabatic cooling, water reuse, etc.

Secondary Water Impact: Thermoelectric power generation is already the largest water consumer in the US.

The Future

In Space and Under the Sea

We're reconsidering the environments computing operates in.

Both extremes are being actively tested.

Space

- Solar energy availability
- Cooling challenges
- Radiation/error tolerance
- Cost of deployment and repairs
- Inter-satellite communication

Sea

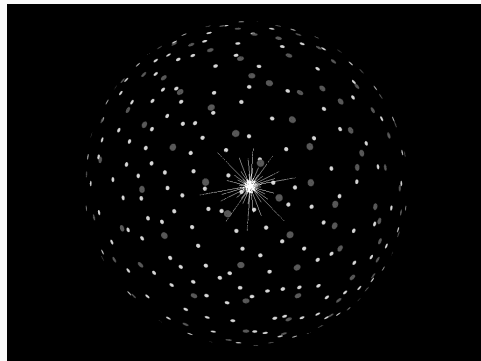
- Water/heat sink availability
- Environmental effects
- Cost of deployment and repairs

Limits of Scale/Computing

What is the limit of computing at scale?

Is there a limit?

Should we self-impose a limit?



Artistic depiction of a Matrioshka Brain

Efficiency and Jevons Paradox

- **The Efficiency Trend (Koomey's Law):**
 - Computations per joule doubled every ≈ 1.57 years for decades (now slower).
 - *Result:* Modern computers are vastly more efficient than their predecessors.
- **Jevons Paradox:**
 - Efficiency improvements increase overall resource consumption..
 - Gains in computing efficiency unlock new applications and demand (e.g. we now use AI for internet searches).

Intrinsic Limits?

- **Lighting Demand:** Capped by biology/physics (e.g., the human eye).
- **Computing Demand:** Capped only by available energy and heat dissipation?

Will gains in computing efficiency lead to reduced resource consumption?

Sufficiency, Sobriety, and Ethics

Sufficiency: What is the least amount of resources/computing to complete a task?

- Example: Do I need a 1 trillion parameter model, or can I use linear regression?
- **Self-sufficiency** example: Solar-powered pocket calculator

Sobriety: Is this a problem I need (computing) to solve?

- Conscious evaluation of tech use. **Re-examine our relationship with technology.**
- Example: Does my toaster need AI to make my morning toast?

Ethics: How do we ensure computing systems at scale are ethically designed, built, operated, used, and decommissioned?

Summary: Computing at Scale

The full stack of computing at scale:

1. **Foundation:** Information processing is valuable, and we use physical computers to perform this task.
2. **Building Blocks:** Transistors drove 50 years of exponential growth (Moore/Dennard), but are hitting lower limits.
3. **New Scales:** We now scale up, out, and across to form massive systems.
4. **Constraints:** This massive scale is colliding with physical realities: energy grids, water supplies, and communities (us).

Our technical ability to scale is **limitless**,
but our resources and ethical boundaries are **limited**.

**What do you want the future
of computing to look like?**